Will It Still Be There? Bike and Dock Availability for Bike Share Toronto

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Project Description

Bike Share Toronto carried 7.8 million rides in 2025 [1]. Riders pick a station based on the current bike count shown in the app. When the app shows a single e-bike, 30 minutes later it is gone 18% of the time in our data, so the rider must detour to another station. Our project aims to reduce unnecessary detours by predicting the probability that a bike or a free dock will still be at a station when the rider arrives, 2–30 minutes later. Building on station-level forecasting [2]–[7], we will (tentatively) train neural networks (an MLP, a per-station GRU/1-D CNN, and a graph neural network over nearby stations [8]–[10]) and keep the one that best beats a gradient-boosted baseline.

Datasets and Number of Samples

Ride data: one row per trip (times, stations, bike ID and model); *station data*: bikes of each type and free docks at every station at one timestamp.

Source	Type	How many (to Oct 16)
Bike Share Toronto ridership, Toronto Open Data [11]	Ride	552,073 trips, Jan–Mar 2026
Public GBFS archive, bike-sharing-history [12]	Station	55,849 snapshots*, Apr 2024 on, ∼16 min apart
Our own 1-min poller of the official GBFS feed [13]	Station	22,466 snapshots*, Sep 29 on, 1 min apart
CityBikes API [14], a mirror of the same feed (fills our gaps)	Station	22,353 snapshots*, overlaps ours

*One snapshot contains all stations (polls in the same minute count once). Stations grew from 792 to 1,075; some snapshots are incomplete.

New Samples and Demonstration

Our pollers add about 1,300 snapshots per day. Because consecutive minutes are nearly identical (99.4% unchanged after 1 min) and the data drifts, we will (tentatively) split by whole days in time order, repeated as data arrives (rolling origin [15], [16]): train on all earlier days, validate on the next week, and test on the week after. The final model is frozen on Nov 22 and tested on later data. Success will be measured in rider terms, such as minutes saved and strandings avoided per trip (fixed after a first benchmark), with staged goals so that every stage yields a usable result. We will demonstrate two applications: (i) a sped-up replay of one busy station through a weekday rush hour on unseen days, showing the app's count, our prediction, and the actual count; and (ii) the bigger picture: time saved and strandings avoided per rider compared with today's app, scaled to all Toronto riders.

Why do you think this project will be successful?

Our data above grows daily, and we have two baselines to beat: the app's own count and our gradient-boosted model.

References

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